# GHK-Cu

Step-by-Step Reconstitution Process

- **Gather supplies:** BAC water, alcohol swabs, insulin syringe, vial
- **Bring BAC water and peptide vial to room temperature before mixing**
- **Removing plastic cap.** Carefully pop off the plastic cap, do not remove metal flange.
- **Clean both vials:** Wipe the rubber stoppers with alcohol swabs
- **Draw BAC water:** Pull your chosen amount into the syringe
- **Inject slowly:** Add the water to the peptide vial, aiming it at the glass wall - not directly onto the powder
- **Swirl gently:** Roll the vial between your palms. Never shake.
- **Wait:** Let it sit 5-10 minutes. Solution should be crystal clear.
- **Store:** Refrigerate immediately. Good for 28-30 days once reconstituted.

**If it's cloudy or won't dissolve:** Let it sit longer. Try gentle warming (under your arm or in lukewarm water). If still cloudy after 30 minutes, contact your source.

## Practical Protocols

The following protocols are not derived from published dose-finding studies. They represent clinical practice patterns and community consensus.

### STANDARD INJECTABLE PROTOCOL (Subcutaneous)

**Dose:** 1 to 2 mg per day

**Frequency:** Daily or 5 days per week

**Duration:** 4 to 8 weeks (can extend to 12 weeks)

**Break:** 4 to 6 weeks between cycles

### RECONSTITUTION (standard for 50 mg vial)

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|---|---|
| **Vial size** | 50 mg |
| **Bac water** | 2.5 to 3 mL |
| **Concentration** | At 2.5 mL, 1 mg = 5 units on insulin syringe |
| **Note** | Solution will turn blue due to the copper. This is normal. |

## Common Questions

**Does the injectable version burn when you inject it?** Yes, injection site burning is one of the most commonly reported experiences with GHK-Cu. The burning can last 15 to 20 minutes. Free copper can temporarily separate from GHK at the injection site, generating mild oxidative activity and histamine release. Strategies to reduce this include diluting the dose with additional bacteriostatic water (add 30 to 90 units of extra water to your drawn dose), rotating injection sites, injecting in fattier areas, and using GHK-Cu blended with BPC-157 (as in the GLOW or KLOW blends). Some people also find that antihistamines taken 30 minutes before injection help.

**Is the topical version effective or do I need the injectable?** Both work, but they serve different purposes. The topical works for skin and hair at the application site. Clinical trials with topical GHK-Cu showed real, measurable improvements in wrinkle reduction, skin density, and collagen production. But the topical will never reach deep tissues or systemic circulation no matter how much you apply. If your goals are limited to facial skin and hair, topical may be sufficient. If you want systemic effects including tissue repair, recovery, and healing throughout your body, the injectable is the route that reaches those targets.

**Does GHK-Cu cause cancer?** No study has demonstrated that GHK-Cu causes cancer. Gene expression research using the Broad Institute's Connectivity Map actually showed that GHK reversed 70% of overexpressed genes in a metastasis-prone colon cancer gene signature and increased expression of numerous caspase, growth regulatory, and DNA repair genes that would be expected to inhibit cancer growth. However, GHK-Cu does promote angiogenesis, which is the same process tumors use to build blood supply. It is contraindicated with active malignancy. No active cancer and no active masses means low theoretical risk. Active tumors or recent cancer history means do not use until cleared by imaging and your oncologist.

**Do I need to cycle GHK-Cu?** For injectable use, yes, cycling is recommended. Standard runs are 4 to 8 weeks on, 4 to 6 weeks off. Some people extend to 12 weeks. For topical use, no cycling is necessary. You can use topical GHK-Cu continuously as part of your skin care routine.

**Can I inject GHK-Cu anywhere or does location matter?** GHK-Cu works systemically once it enters your bloodstream, so subcutaneous injection anywhere works for general anti-aging and systemic benefits. For a specific injury or localized goal (like skin tightening in a particular area), injecting closer to the target tissue gives higher local concentration before the peptide clears due to its short half-life.

**Why did my GHK-Cu solution turn blue?** The blue or blue-green color comes from the copper in the compound. This is completely normal and expected. If your reconstituted GHK-Cu is not slightly blue, that would actually be more concerning.

**Is injectable GHK-Cu more effective than topical for skin tightening after weight loss?** Injectable is going to be more effective than topical for skin tightening because it reaches systemic circulation. The topical versions do not penetrate deep enough to impact collagen synthesis at the level needed for significant skin laxity. For skin laxity after weight loss, a standard injectable protocol of 1 to 2 mg daily for 8 to 12 weeks is a reasonable starting point.

## Side Effects

### In Published Research

Clinical studies of topical GHK-Cu report an excellent safety profile with decades of use. The most common adverse events in a 2023 safety review analyzing 12 studies (N=512) were transient erythema (4.2%) and pruritus (2.8%), with no systemic effects consistently reported. The 12-week clinical trial with 71 women reported no adverse events.

### What Users Report

**Common (with injectable use).** Burning or stinging sensation at the injection site lasting 15 to 20 minutes. Redness, swelling, or a bump at the injection site that resembles hives. Temporary blue or green discoloration at the injection site from the copper. These reactions occur because free copper can temporarily separate from GHK at the injection site, generating mild oxidative activity and histamine release. This is the most universally reported experience with injectable GHK-Cu.

**Less common.** Metallic taste in the mouth (temporary). Mild changes in appetite or energy levels. Headache. Fatigue.

**Rare.** Some users report a phenomenon called "copper uglies" where skin quality temporarily worsens (increased wrinkling or sagging) before improving. This appears more associated with topical use at high concentrations than with injectable use. Reports exist on Reddit and skin forums, though some dermatologists dispute whether this is a real effect of GHK-Cu or caused by other factors.

Most side effects are dose-dependent and temporary, typically resolving within hours to days.